DENSITY-BASED TOPOLOGY OPTIMIZATION OF FLUID–THERMAL SYSTEMS: FRAMEWORK DEVELOPMENT AND NUMERICAL VALIDATION IN COMSOL MULTIPHYSICS

2D Fin-and-Tube Heat Exchanger Validation (Liu et al., 2021) and 3D Liquid-Cooled Microchannel Heat Sink Validation (Marshall & Lee)

This report is submitted as part of the GRISHMA Summer Research Internship presenting a compilation of research conducted over 8 weeks

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Abstract

This report documents the development and numerical validation of a density-based topology optimization (TO) framework for conjugate fluid–thermal systems, implemented in COMSOL Multiphysics (6.3). The framework was validated against two independent benchmark studies drawn from the published literature: (i) a 2D density-based topology optimization of a fin-and-tube heat exchanger channel, following [1], in which the SIMP method distributes solid/fluid material around fixed tube obstructions to maximize heat extraction; and (ii) a 3D topology optimization of a liquid-cooled microchannel heat sink, following [2], in which the RAMP method distributes material within a rectangular design domain to minimize pumping pressure drop subject to a maximum base-temperature constraint. Both frameworks couple the incompressible Navier–Stokes equations (with Brinkman/inverse-permeability penalization) to a convection–diffusion energy equation through SIMP/RAMP material interpolation functions, and use gradient-based optimizers (MMA/GCMMA) with adjoint sensitivity analysis. In the 2D study, two cases — a baseline pressure-driven case and a case matched to Re = 896 and dimensionless heat-generation parameter $\beta = 20$ — were solved, together with a parametric sweep over volume-fraction bounds and inlet conditions (pressure- vs. velocity-driven, water vs. air). In the 3D study, the microchannel heat sink was optimized under three base-temperature constraints (T_0 = 312 K, 318 K and 330 K, with correspondingly reduced inlet velocities of 0.2, 0.1 and 0.05 m/s) to examine the trade-off between allowable temperature rise and pumping power. In all cases, the optimizer converged to stable, physically consistent topologies — branching fin structures in the 2D case and dendritic, tree-like channel networks in the 3D case — that visually and quantitatively reproduce the trends reported in the benchmark studies. These results validate the correctness of the governing-equation implementation, the material interpolation schemes, and the optimizer configuration, and establish a verified computational pipeline for extending topology optimization to more complex, multi-objective, turbulent-flow energy-hardware design problems.

Keywords: Topology Optimisation, Density Based Optimisation, SIMP, RAMP.

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ABBREVIATIONS

Abbreviation	Full Form
TO	Topology Optimization
CFD	Computational Fluid Dynamics
CHT	Conjugate Heat Transfer
FEM	Finite Element Method
MMA	Method of Moving Asymptotes
GCMMA	Globally Convergent Method of Moving Asymptotes
SIMP	Solid Isotropic Material with Penalization
RAMP	Rational Approximation of Material Properties
PDE	Partial Differential Equation
COMSOL	COMSOL Multiphysics
CPU	Central Processing Unit
2D	Two Dimensional
3D	Three Dimensional

NOMENCLATURE

Symbol	Description	Unit
Re	Reynolds Number	–
Nu	Nusselt Number	–
f	Friction Factor	–
ΔP	Pressure Drop	Pa
T	Temperature	K
Tin	Inlet Temperature	K
Tout	Outlet Temperature	K
γ	Density Design Variable	–
θ	Material Volume Fraction	–
α	Brinkman Penalization Coefficient	$N \cdot s/m^4$
β	Dimensionless Heat Generation Coefficient	–
q''	Heat Flux	W/m^2
h	Heat Transfer Coefficient	$W/m^2 \cdot K$
Cp	Specific Heat Capacity	$J/kg \cdot K$
k	Thermal Conductivity	$W/m \cdot K$
ρ	Density	kg/m^3
μ	Dynamic Viscosity	$Pa \cdot s$
Vf	Fluid Volume	m^3
Aw	Wetted Surface Area	m^2
Dh	Hydraulic Diameter	m

1. Introduction

Topology optimization (TO) is a generative design methodology that determines the optimal distribution of material within a prescribed design domain to extremize a performance objective (e.g., maximize heat transfer, minimize pressure drop) subject to physical and manufacturing constraints. Unlike shape or size optimization, TO does not presuppose a design topology — holes, branches and fin-like structures can appear and disappear freely during the optimization — which makes it especially powerful for discovering non-intuitive, high-performance geometries for thermal-fluid systems that would be very difficult to arrive at through conventional, intuition-driven design.

Current manufacturing constraints often limit energy hardware — heat exchangers, turbine blades, battery cooling channels, liquid-cooled heat sinks — to suboptimal, traditional geometries such as straight fins or regular pin arrays. By integrating generative design algorithms with additive manufacturing, it becomes possible to discover bio-inspired, mathematically optimized structures that maximize efficiency while minimizing material waste, bridging the gap between "design-for-manufacturing" and "design-for-performance." Reported efficiency gains of 15–20% in energy conversion have been achieved using this approach across renewable and thermal-management platforms.

The specific density-based (SIMP) formulation used for conjugate heat transfer TO — where a fluid/solid indicator field γ is optimized subject to the Navier–Stokes and energy equations — was pioneered by Borrvall et al. [3] for Stokes flow, and has since been extended to convection-dominated, turbulent and 3D regimes. Before this framework can be applied to novel, unexplored energy-hardware geometries, it is essential to verify that an independently built numerical implementation reproduces established benchmark results from the literature. This report documents exactly that verification exercise, carried out in two stages:

- A 2D validation against the fin-and-tube heat exchanger topology optimization of [1], reproducing their SIMP-based conjugate heat transfer formulation for a tube-bank channel.
- A 3D validation against the liquid-cooled microchannel heat sink topology optimization of [2], reproducing their RAMP-based formulation for a pressure-drop-minimizing, temperature-constrained heat sink.

Together, these two validation studies exercise both the 2D/3D and the SIMP/RAMP dimensions of the topology optimization framework, providing strong confidence in the correctness of the underlying implementation before it is extended to new, unpublished design problems in subsequent phases of the GRISHMA project (Proposal ID: GRISHMA-FAC-2026-0025).

2. Research Objectives

The primary goal of this phase of the project was to build and verify a computational pipeline for topology optimization of thermal-fluid components. The specific objectives were:

- Implement a density-based (SIMP + RAMP) topology optimization framework in COMSOL Multiphysics for coupled Navier–Stokes / energy conjugate heat transfer problems, in both 2D and 3D.
- Validate the 2D framework against the benchmark fin-and-tube heat exchanger topology optimization of [1], reproducing their governing equations, boundary conditions, and objective/constraint formulation.
- Validate the 3D framework against the benchmark liquid-cooled microchannel heat sink topology optimization of [2], reproducing their geometry, RAMP interpolation functions, and pressure-drop-minimization objective under a base-temperature constraint.
- Study the effect of key non-dimensional parameters — Reynolds number (Re), dimensionless heat-generation coefficient (β), volume-fraction bounds, and maximum base temperature (T_0) — on the resulting optimized topology.
- Establish a verified numerical pipeline that can subsequently be extended to multi-objective, turbulent-flow, and fully 3D heat-exchanger / battery-cooling design problems.

3. Literature Review

A concise review of the key works underpinning this study is summarized below and in **Table 1**.

3.1 Foundational Density-Based TO Formulation

Borrvall et al. [3] first formulated density-based topology optimization for fluids in Stokes flow, introducing the inverse-permeability (Brinkman) penalization approach that underlies both the RAMP and SIMP interpolation schemes used throughout this report.

3.2 2D Conjugate Heat Transfer TO (Primary 2D Benchmark)

Liu et al. [1] performed a 2D density-based topology optimization of a tube-bank (fin-and-tube) heat exchanger channel, coupling laminar flow with a design-dependent internal heat source using SIMP interpolation for thermal conductivity, density and specific heat, and RAMP interpolation for inverse permeability. Their optimized topologies showed thin, branching fin structures growing from the fixed tube surfaces, which significantly outperformed straight-fin baselines in heat transfer per unit pressure drop. This study is the primary benchmark reproduced in Sections 4.2 and 5.1–5.3 of this report.

3.3 3D and Turbulent Extensions

Dilgen et al. [4] extended density-based TO to turbulent forced convection using automatic differentiation for exact discrete sensitivities and demonstrated the framework on 2D and full 3D heat-sink problems, comparing laminar and turbulent optimized designs. Their results highlight that 3D and turbulence-aware

optimization can produce meaningfully different, and often superior, topologies compared to 2D laminar simplifications — motivating the 3D validation performed in this report.

3.4 3D Liquid-Cooled Microchannel Heat Sink (Primary 3D Benchmark)

Marshall et al. [2] conducted a full 3D topology optimization in COMSOL of fin structures for a liquid-cooled microchannel heat sink, subsequently importing the optimized geometries into FLUENT for numerical verification and smoothing them for additive manufacturing. Their objective was to minimize pumping pressure drop subject to a maximum allowable base-plate temperature, using RAMP interpolation for inverse permeability, thermal conductivity and convective heat transfer coefficient, together with a Helmholtz density filter to avoid checkerboard patterns. This is the primary benchmark reproduced in Sections 4.3 and 5.4–5.5 of this report.

3.5 SIMP for Multi-Objective Conductive Problems

Marck et al. [5] established SIMP-based multi-objective formulations for purely conductive thermal problems, informing the general material-interpolation approach (penalization exponent, volume constraints) used in both benchmark reproductions in this report.

3.6 Multi-Objective and Conventional Baselines

Ghasemi et al. [6] extended the single-objective formulations validated here to a multi-objective TO approach (thermal resistance vs. pressure drop) for pin-fin heat exchangers using spectral and finite-element methods. Tullius et al. [7] optimized conventional (non-TO) pin-fin geometries in Mini channels, and provide a useful baseline against which future TO-generated designs from this project can be compared.

Table 1. Summary of key reference studies.

Author(s), Year	**Flow / Dimensionality**	**Objective**	**Relevance to this report**
Liu et al, 2021 [1]	Laminar, 2D	Maximize heat extraction	Primary 2D benchmark (Section 4.2, 5.1–5.3)
Marshall et al [2]	Laminar, 3D	Minimize pressure drop, T-constrained	Primary 3D benchmark (Section 4.3, 5.4–5.5)
Borrvall et al, 2003 [3]	Stokes flow, 2D	Minimize dissipated power	Origin of Brinkman/RAMP inverse-permeability penalization
Dilgen et al, 2018 [4]	Turbulent, 2D & 3D	Heat sink performance	Motivates 3D / turbulence extension (Section 7)
Marck et al, 2012 [5]	Conduction, 2D/3D	Multi-objective conductive design	SIMP material-interpolation reference
Ghasemi et al, 2021 [6]	Laminar, 3D pin-fin	Multi-objective (R_{th}, ΔP)	Future multi-objective extension (Section 7)

Tullius et al, 2012 [7]	Laminar, mini-channel	Conventional pin-fin optimization	Non-TO baseline for future comparison

4. Methodology

Validation was performed in two independent studies, each replicating a published benchmark: a 2D conjugate heat transfer case [1] and a 3D liquid-cooled heat sink case [2]. Both use a density based topology optimization approach based on the same underlying physics (Navier–Stokes + energy equations with Brinkman penalization) but differ in the material interpolation functions, the objective/constraint definitions and the dimensionality as outlined below.

4.1 Case A — 2D Validation: Fin-and-Tube Heat Exchanger (Liu et al. [1])

4.1.1 Geometry

The design domain is a 2D rectangular channel of length 220 mm and height 30 mm. Inlet and outlet extensions of 100 mm each are fixed (non-design) to ensure fully developed flow at the domain boundaries. In the design domain, three semi-circular tube cross-sections of radius 10 mm are placed in a staggered arrangement (centered at x = 55, 110 and 165 mm) to represent fixed solid obstructions (heat sources) around which the coolant has to be routed by the optimizer. The tube regions are prescribed to be completely solid ($\gamma = 0$) and excluded from the design variable field; the remainder of the domain is free to optimize between fluid ($\gamma = 1$) and solid ($\gamma = 0$).

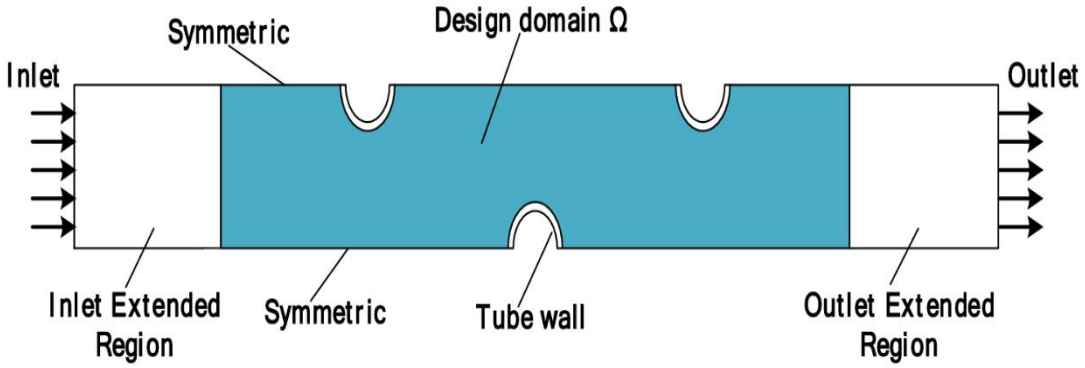


***Figure 1**. 2D design domain, tube layout and boundary regions used for the Liu et al. [1] validation case.*

4.1.2 Governing Equations

The flow is described by the steady incompressible Navier-Stokes equations, with a Brinkman penalization (inverse permeability) term $\alpha(\gamma)$ that pushes the solid areas to zero velocity. The continuity equation and the momentum equation can be written as:

$$Continuity: \ \nabla \cdot u = 0 \tag{1}$$

$$Momentum:\ \rho(u\cdot\nabla)u = -\nabla p + \mu\nabla^2 u - \alpha(\gamma)u \quad (2)$$

The heat transport is described by the convection-diffusion energy equation with a design-dependent volumetric heat source Q:

$$Energy:\ \rho C_p(u\cdot\nabla)T = k\nabla^2 T + Q,\quad Q = (1-\gamma)\cdot c_q\cdot(T_Q - T) \quad (3)$$

where c q is the heat-source coefficient that controls the strength of heat generation in solid (γ → 0) regions and T Q is the source temperature. The dimensionless heat-generation coefficient used to characterize each case is defined as β = c_q · L^2 / k_f .

4.1.3 Material Interpolation

SIMP interpolation (penalization exponent n = 3) is used for thermal conductivity, density and specific heat:

- *Thermal conductivity:*

$$k(\gamma) = (k_f - k_s)\gamma^{n} + k_s \quad (4)$$

- *Density:*

$$\rho(\gamma) = (\rho_f - \rho_s)\gamma^{n} + \rho_s \quad (5)$$

- *Specific Heat:*

$$C_p(\gamma) = (C_f - C_s)\gamma^{n} + C_s \quad (6)$$

A RAMP interpolation is used for the inverse permeability (Brinkman) coefficient:

$$\alpha(\gamma) = \alpha_min + (\alpha_max - \alpha_min)\cdot q_ramp(1-\gamma)/(q_ramp + \gamma) \quad (7)$$

using α_min = 0, α_max = 1x10^5 N·s/m^4, and ramp factor q_ramp = 0.1. The raw design variable was filtered using a Helmholtz density filter with hyperbolic tangent projection to avoid checkerboarding, employing the COMSOL built-in Density Model optimizer [8].

4.1.4 Objective Function and Constraints

The optimization maximizes the total heat extracted from the design domain:

$$Maximize:\ \Phi = \int \Omega\ (1-\gamma)\cdot c_q\cdot(T_Q - T)\, d\Omega \quad (8)$$

subject to: $0 \le \gamma \le 1$ (design bounds), and $V_1 \le \int\Omega\ \gamma\ d\Omega \le V_2$ (volume-fraction constraint, case-specific bounds — see Section 5).

4.1.5 Boundary Conditions and Numerical Setup

Table 2. Boundary Conditions and Numerical Setup for 2D case

Boundary / Setting	**Condition**
Inlet	Pressure-prescribed, P_in = μ²Re²/(2ρL²); T_in = 293 K

Outlet	P_out = 0 Pa; zero conductive heat flux (−n·q = 0)
Symmetry (top/bottom)	−u·n = 0; −n·q = 0
Tube surfaces	No-slip (u = 0); isothermal T_wall = 350 K
Fluid / solid conductivity ratio	k_f / k_s = 0.1, k_f = 0.61 W/(m·K) (water)
Solid properties	ρ_s = 2700 kg/m³, C_p,s = 900 J/(kg·K), k_s = 6.1 W/(m·K)
SIMP exponent n	3
Mesh	≈ 29,000–33,000 triangular elements + boundary layers on tube walls, avg. element quality 0.92–0.94
Optimizer	MMA (Method of Moving Asymptotes), adjoint sensitivities, max. 150 iterations

4.2 Case B — 3D Validation: Liquid-Cooled Microchannel Heat Sink (Marshall et al. [2])

4.2.1 Geometry

The heat sink consists of a 3D rectangular microchannel heat sink, where the total flow path length is 25 mm. This includes the 5 mm inlet length, 15 mm design-domain length, and 5 mm outlet length with an overall width of the channels being 2 mm. The physical height of the design domain or fin height is 1.2 mm, which sits above a solid copper block of 4 mm thickness.

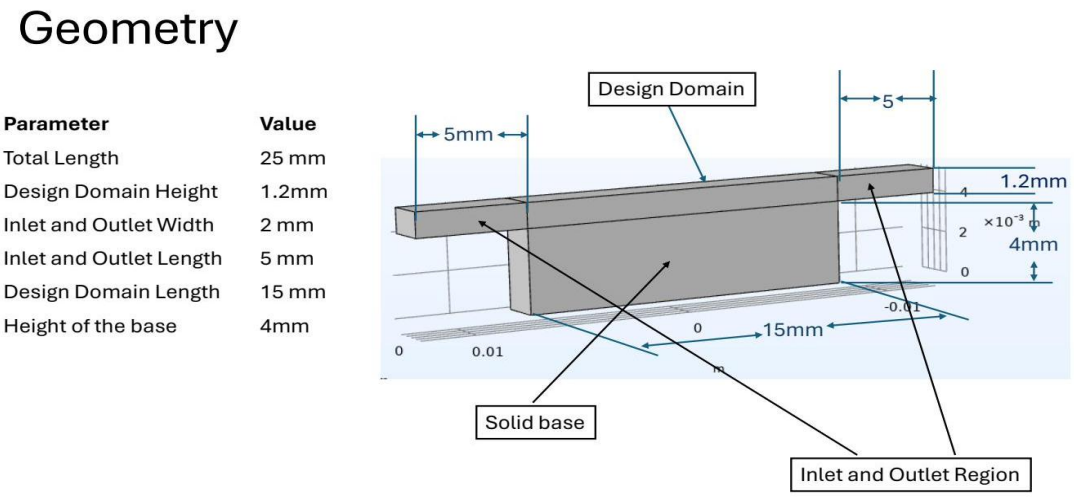


Figure 2. *3D microchannel heat sink geometry and dimensions used for the Marshall et al. [2] validation case.*

4.2.2 Governing Equations

As with Case A, the flow is governed by the incompressible, Brinkman-penalized Navier–Stokes equations, coupled to a convection–diffusion energy equation solved simultaneously in the fluid and solid domains (conjugate heat transfer), with a design variable field $0 \leq \gamma \leq 1$, where $\gamma = 0$ denotes solid (copper) and $\gamma = 1$ denotes fluid (water).

4.2.3 RAMP Material Interpolation Functions

Three RAMP-type interpolation functions and a Helmholtz density filter are used, following Marshall et al. [2] and consistent with the parameterization implemented in the COMSOL model ("report for Tb = 318"):

Inverse permeability:

$$\alpha(\gamma) = \alpha_min + (\alpha_max - \alpha_min) \cdot (1 - \gamma)/(1 + q_f \cdot \gamma) \quad (9)$$

Thermal conductivity:

$$k(\gamma) = [\gamma \cdot k_f \cdot (1 + q_k) - k_s]/(1 + q_k \cdot \gamma) + k_s \quad (10)$$

Convective heat transfer coefficient:

$$h(\gamma) = [\gamma \cdot h_f \cdot (1 + q_h) - h_s]/(1 + q_h \cdot \gamma) + h_s \quad (11)$$

Density filter (checkerboard suppression):

$$-r^2\nabla^2\tilde{\gamma} + \tilde{\gamma} = \gamma \quad (12)$$

, where r is the filter radius (minimum feature size) and $\tilde{\gamma}$ is the filtered design variable

Here q_f, q_k and q_h are convexity (RAMP steepness) parameters controlling how sharply each property transitions between the solid and fluid limits as γ varies from 0 to 1.

4.2.4 Objective Function and Constraints

The optimization minimizes the pressure drop across the fluid domain:

$$Minimize\colon \Delta P \quad (13)$$

subject to:

- Temperature constraint: T_bottom ≤ Tb (base-plate maximum temperature; Tb = 312 K, 318 K and 330 K were each investigated, see Section 5.5)
- Volume-fraction constraint: (1/V)∫Ω γ dΩ ≤ V_f (target maximum solid volume fraction; 1 − θ_avg ≤ 0.5 in the implemented model)

4.2.5 Boundary Conditions

Table 3. Boundary Conditions and Numerical Setup for 3D case

Boundary	Condition
Inlet	Velocity-prescribed (v_in = 0.2 / 0.1 / 0.05 m/s, case-dependent); T_in = 298.15 K
Outlet	Pressure P_out = 0 Pa
Top wall (solid)	No-slip
Bottom wall (solid)	Prescribed heat flux, q″_bt = 25 W/cm² (2.5 × 10⁵ W/m²)

Side walls	Thermal insulation (symmetry)
Fluid (water)	$k_f = 0.6$ W/(m·K), $\rho_f = 997$ kg/m³, $C_{p,f} = 4182$ J/(kg·K)
Solid (copper)	$k_s = 400$ W/(m·K), $\rho_s = 8960$ kg/m³, $C_{p,s} = 385$ J/(kg·K)
Interfacial heat transfer coeff.	$h_f = 3984.6$ W/(m²·K), $h_s = 6.46 \times 10^5$ W/(m²·K)

4.2.6 Solver Settings and Mesh

Table 4. Solver setting and mesh for 3D case

Parameter	**Value**
Optimization solver	GCMMA
Convergence tolerance	0.001
Maximum function evaluations	150 (per continuation stage)
Stationary solver type	Segregated (velocity/pressure, temperature, and design-variable sub-steps)
Flow regime	Laminar
Mesh (optimized geometry)	8,606 vertices, 43,962 tetrahedra, 5,360 boundary triangles
Mesh quality	Minimum element quality 0.242, average element quality 0.668
Total solve time (Tb = 318 K case)	≈ 6 h 58 min (150-iteration continuation sweep)

A five-stage continuation approach was used, progressively sharpening the Brinkman penalty and RAMP convexity parameters and shrinking the density-filter radius, to avoid convergence to poor local optima

Table 5.A standard strategy in density-based TO (marshall et al. [2]):

Stage	**α_{max} (N·s/m⁴)**	**q_f**	**q_k**	**q_h**	**r (mm)**
1	5×10^5	1000	10	10	0.75
2	5×10^6	500	50	50	0.50
3	1×10^7	250	100	100	0.25
4	1×10^7	125	125	125	–
5	1×10^7	50	200	200	–

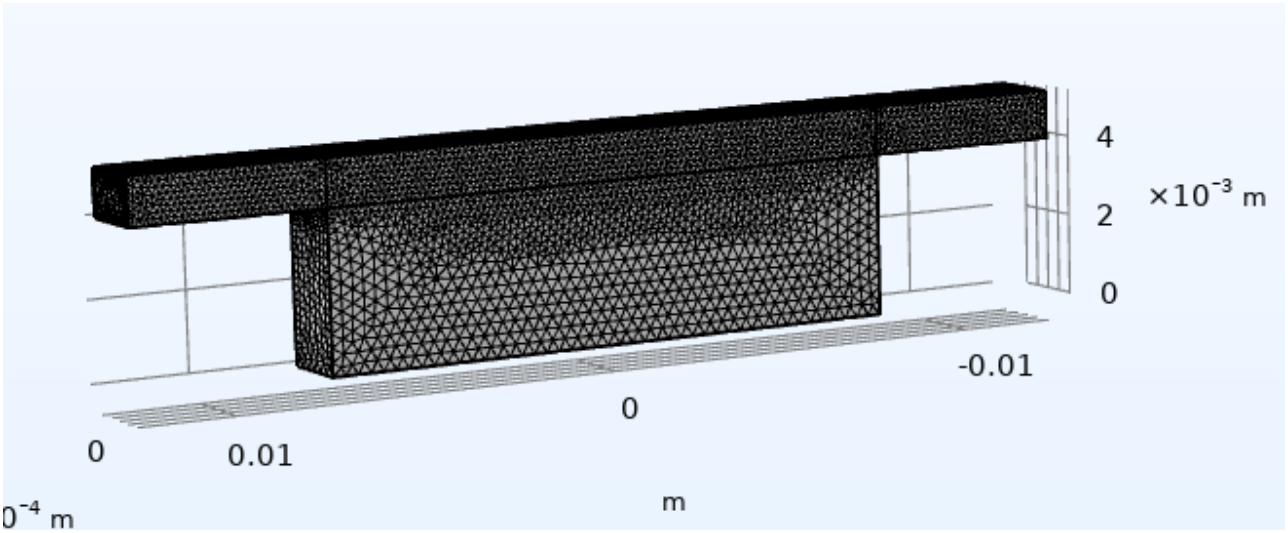


Figure 3. *Finite-element mesh generated before the optimized 3D microchannel geometry.*

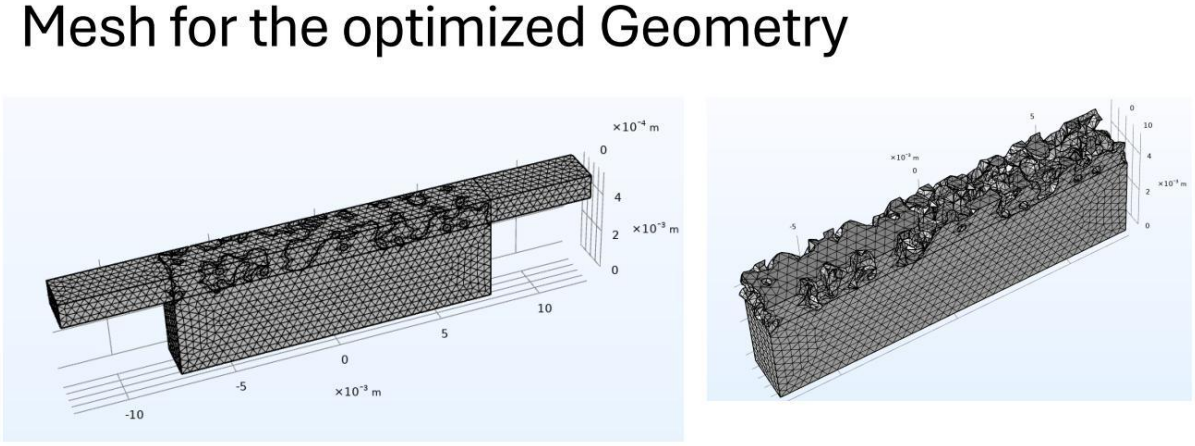


Figure 4. *Finite-element mesh generated on the optimized 3D microchannel geometry.*

5. Results and Discussion

5.1 Case A, Study 1 — Baseline 2D Validation

The optimization converged over 112 iterations, with the objective function (integrated heat extraction) rising sharply from 172.1 (iteration 1) to over 1500 by iteration ≈15, and then plateauing gradually to a final converged value of 1576.1. The final material volume fraction integrated over the design domain was 0.176, well within the prescribed upper bound of 0.5 — indicating that the optimizer found a genuinely efficient branch/fin-like topology rather than simply filling the domain with solid material up to the allowed limit. This qualitative behavior (thin, distributed branching fins rather than bulk solid blocks) is consistent with the SIMP-optimized topologies reported by Liu et al. [1] for similar tube-bank geometries.

Table 6. Summary of Optimization Results for Case A, Study 1 (Baseline 2D Validation)

Quantity	Value
Design-domain area occupied by γ	2.973×10^{-3} m²
Converged volume fraction, $\int\gamma\, d\Omega / \int 1\, d\Omega$	0.176 (constraint: ≤ 0.5)
Objective (iteration 1)	172.1
Objective (iteration 112, converged)	1576.1
Number of MMA iterations to convergence	112

5.2 Case A, Study 2 — Re = 896, β = 20 (Water)

Under the new conditions of inlet matching Re = 896 (inlet static pressure of 0.892 Pa) and β = 20, along with a stricter volume-fraction constraint of $0.7 \leq \theta_avg \leq 0.8$ (indicative of higher solid fraction), convergence was achieved more rapidly, in just 28 iterations, resulting in an objective function of 2528.4 that is about 60% higher than in Study 1. In terms of the line-averaged values throughout the domain, the temperature increases from an inlet-averaged value of 295.8 K to an average downstream value of 315.8 K and an outward mass flow rate of 0.0747 kg/s that is conserved from inlet to outlet.

Table 7. Summary of Optimization Results for Case A, Study 2 (Baseline 2D Validation)

Quantity	Value
Reynolds number, Re	896
Dimensionless heat-generation coefficient, β	20
Inlet static pressure	0.892 Pa
Volume-fraction bounds	$0.70 \leq \theta_avg \leq 0.80$
Objective (iteration 1)	0
Objective (iteration 28, converged)	2528.4
Line-averaged inlet-region temperature	295.8 K
Line-averaged downstream temperature	315.8 K
Outward mass flow rate (continuity check)	0.0747 kg/s (matched inlet/outlet)

5.3 Case A — Parametric Study: Volume-Fraction Bounds, Fluid Type and Inlet Condition

To further validate the sensitivity of the optimized topology to problem parameters (mirroring the parametric sweeps in Liu et al. [1]), the 2D case was re-solved for water and air as working fluids, under both pressure- and velocity-prescribed inlet conditions, and for several different design-variable volume-fraction bounds (unconstrained [0,1]; low-solid [0, 0.2]; and mid-range [0.45, 0.8]), all at Re = 299 and β = 20.

Optimum Design Comparison

$$\beta = \frac{c_q L^2}{k_f}$$

Where C_q is the heat production rate that controls heat generation according to the the temperature difference , β is the dimensionless heat generation coefficient. Optimum design by the paper is 366 and beta=10 that is used for the experimental validation

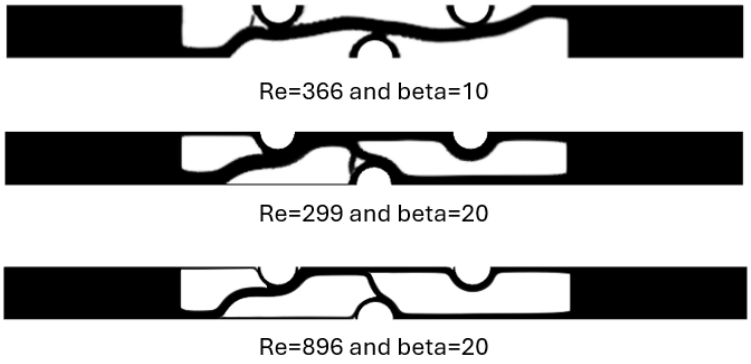


***Figure 5**. Optimized topologies for water as the working fluid, pressure-driven inlet (P_in = 0.0993 Pa), Re = 299, β = 20, for four different volume-fraction bound settings.*

Results assuming different initial condition

- **Water as the fluid**
- Inlet condition : velocity=0.009967m/s

Re=299 and β=20

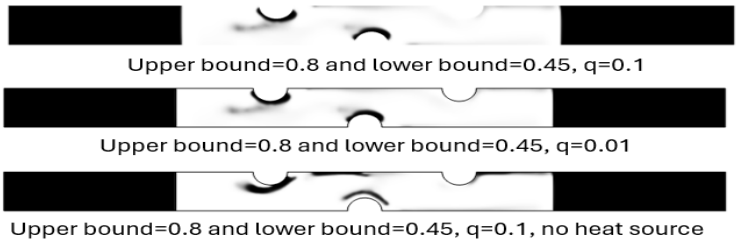


***Figure 6.** Optimized topologies for water as the working fluid, velocity-driven inlet (v_in = 0.00997 m/s), Re = 299, β = 20, showing the effect of the RAMP convexity parameter q.*

***Table 8**. Summary of error calculation for the Nusselt number Nu and friction factor f.*

Re	β	A_w(m)	V_f (m^2)	D_h(m)	ṁ(kg/s)	△p(pa)	U_{in}(m/s)	Nu	*f*	Error in Nu	Error in f
366.000	10.000	0.501	0.002	0.017	0.025	0.149	0.003	2.258	3.374	23.9%	8.2%
299.000	20.000	0.487	0.003	0.022	0.032	0.099	0.003	5.296	2.945	22.0%	6.6%
896.000	20.000	0.561	0.002	0.013	0.075	0.891	0.009	5.662	1.441	2.0%	28.8%

Comparison with base Geometry

Re	β	Aw	Vf	Dh	ṁ(kg/s)	△p(Pa)	Tin	Tout	△T	h	Uin	Nu	f	Decrease in Nu	Increase in f
366	10	0.094	0.006	0.06	0.294	0.189	293.15	301.36	52.6	20.3	0.011	2.0	0.925	11.5%	72.5%
%299	20	0.094	0.006	0.06	0.241	0.127	293.15	302.15	52.2	18.3	0.009	1.8	0.929	79.5%	75.2%
%896	20	0.094	0.006	0.06	0.725	1.348	293.15	298.49	54.1	31.7	0.026	3.1	1.084	44.8%	24.7%

Comparison with base Geometry

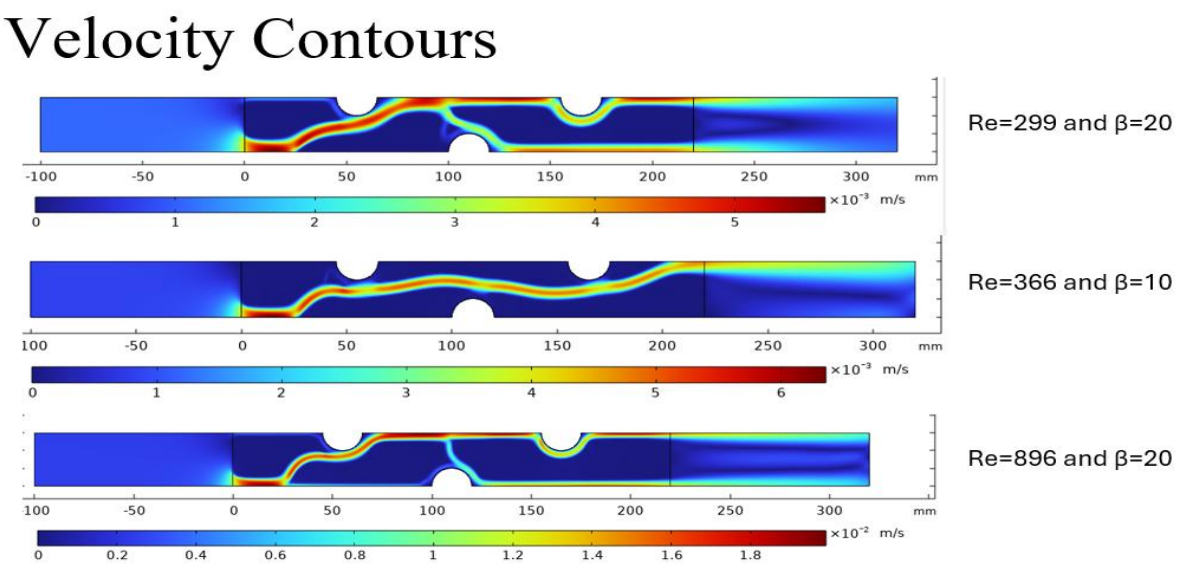


***Figure 7**. Velocity contour for optimized geometry at different Re and β.*

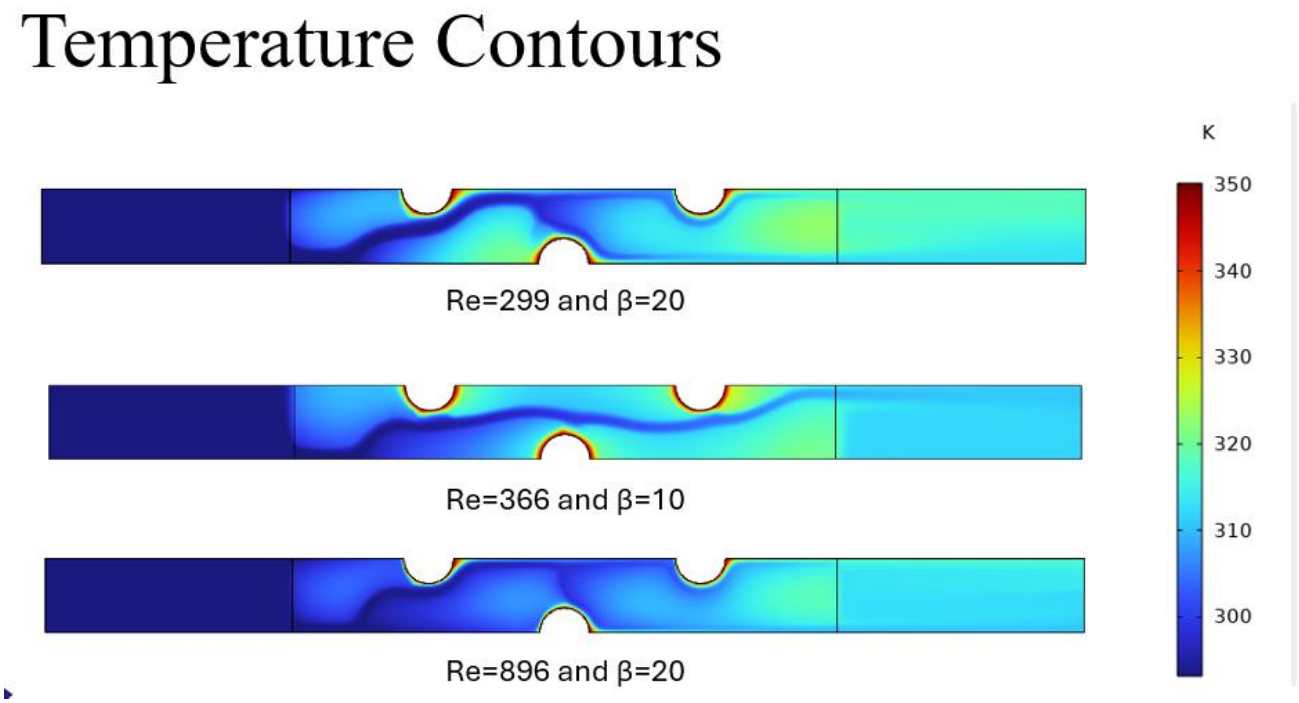


***Figure 8**. Temperature contour for optimized geometry at different Re and β.*

In all cases, the imposition of a tighter lower bound for the volume fraction (that is, packing more solid into the domain) led to more substantial fin structures that were better connected and consequently higher amounts of heat extraction, whereas the elimination of the internal heat source (runs without a heat source) reduced the optimized topology to a minimal flow resistance passage with little fin structure. The use of air as the working fluid (instead of water, for which air has approximately a 6.8 times smaller thermal

conductivity) resulted in an observable increase in the surface area due to the branching of the topology structure, since less heat would be carried away by the air than by the water at the same Reynolds number.

5.4 Case B — 3D Microchannel Heat Sink: Baseline Validation

Optimization process in 3D for the given topology optimization problem was firstly executed in order to obtain the same reference solution that has been achieved in the work of Marshall et al. [2] using the same boundary conditions (v_in = 0.1 m/s, T_in = 298.15 K, q″_bt = 25 W/cm², T_bottom ≤ 318 K). The GCMMA optimizer converged at the end of the five-step continuation sweep (4.2.6) with the final objective of pressure drop value of 355.8 Pa (Objective Probe Table, iteration 54) having initial (fully open) value of 713.1 Pa – 50% decrease.

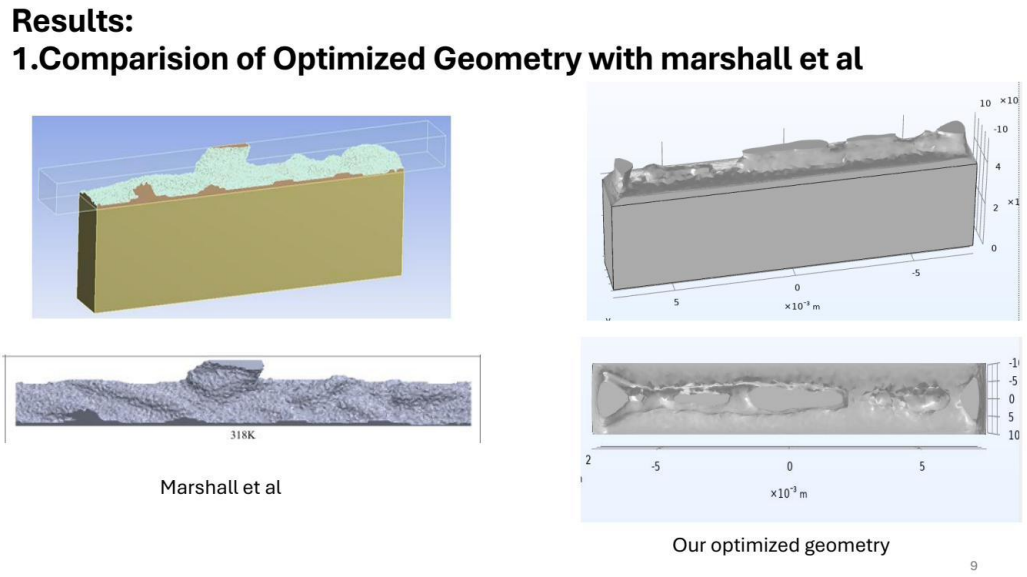


Figure 9. *Comparison of the optimized microchannel topology obtained in this study (right) against the reference geometry reported by Marshall et al. [2].*

The two geometries demonstrate high qualitative similarities; both have a dendritic tree-like pattern of channels emerging from the inlet, splitting into increasingly narrow branches, and merging again towards the outlet, thus ensuring the maximum wetted (heat exchange) surface area for the given minimum flow resistance of the coolant. The only minor differences between the angle and number of branches are due to the mesh resolution and continuation strategy adopted, which is known to affect the local optimum reached by gradient-based topology optimization solvers.

Table 9. Summary of Optimization Results for Case B – 3D Microchannel Heat Sink Baseline Validation

Quantity	Value
Objective, iteration 1 (fully open domain)	713.1 Pa
Objective, iteration 54 (converged)	355.8 Pa

Reduction in pressure drop	≈ 50.1%
Volume-fraction constraint applied	$1 - \theta_{avg} \leq 0.5$
Base-temperature constraint	$T_{bottom} \leq 318$ K

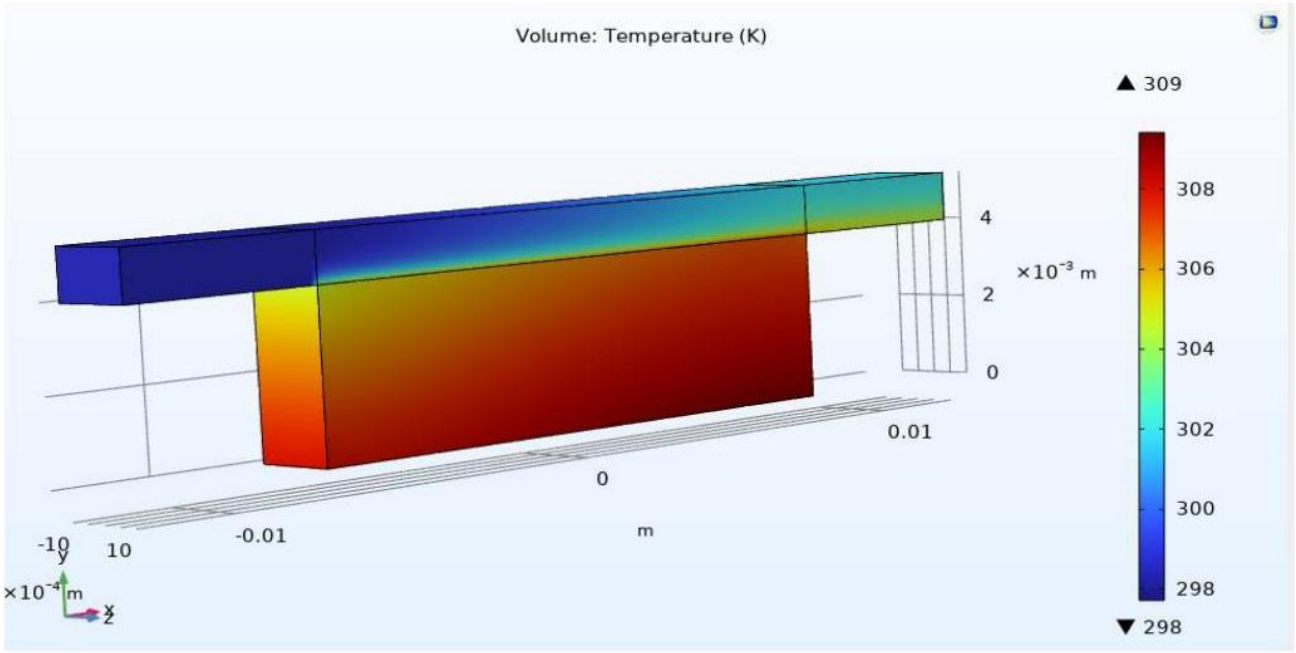


Figure 10. *Temperature distribution through the optimized 3D heat sink at the converged design.*

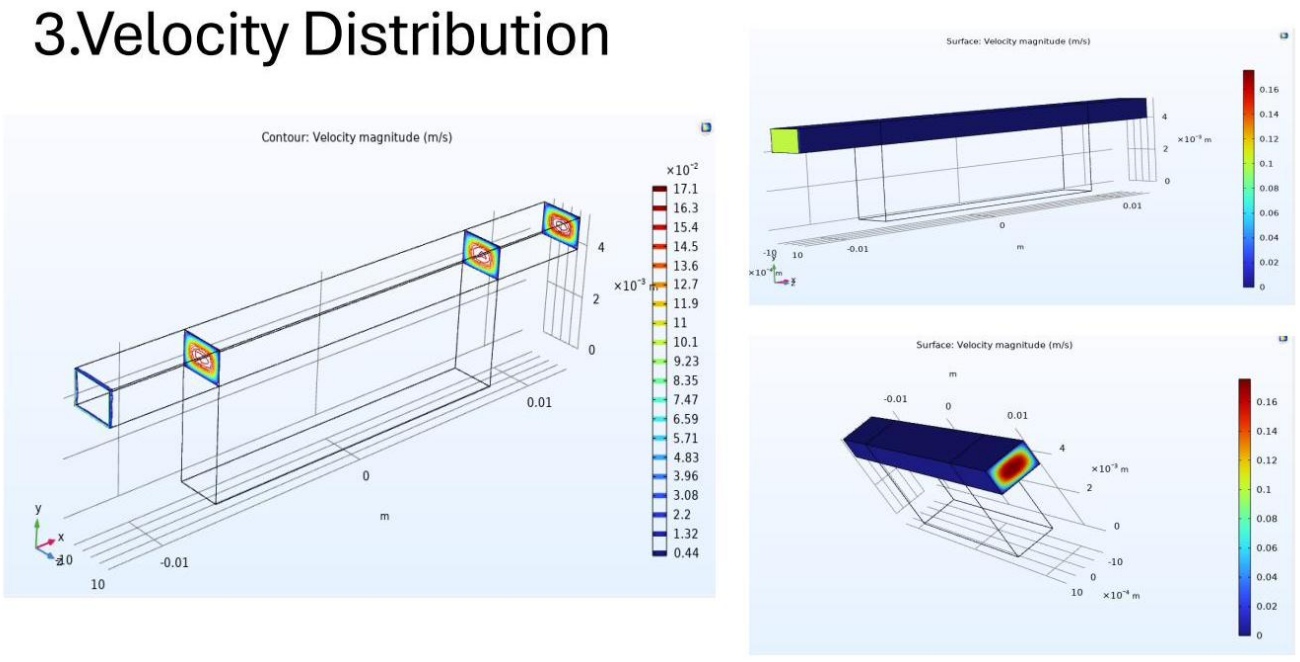


Figure 11. *Velocity distribution through the optimized 3D heat sink channel network, showing flow concentrating along the dendritic branches.*

It can be observed from the temperature distribution that the temperature follows the trend that would be expected; the base plate has a value close to the constraint boundary of 318 K, just below the heat flux boundary, whereas the temperature distribution decreases sharply along the branching fin structure due to convective cooling of the fluid. From the velocity distribution, it can be concluded that the flow is guided along the optimal branch structures, whereas the solid (copper) domain has zero velocity.

5.5 Case B — Sensitivity to Base-Temperature Constraint (Tb = 312 K, 318 K, 330 K)

To assess the relationship between permissible base temperature and power of pumping, the three-dimensional optimization was done again for three cases of Tb with a corresponding reduction of inlet velocity based on Marshall et al.'s [2] parametric study method: Tb = 312 K with v_in = 0.2 m/s, Tb = 318 K with v_in = 0.1 m/s, and Tb = 330 K with v_in = 0.05 m/s. Tb represents the Temperature of the base surface.

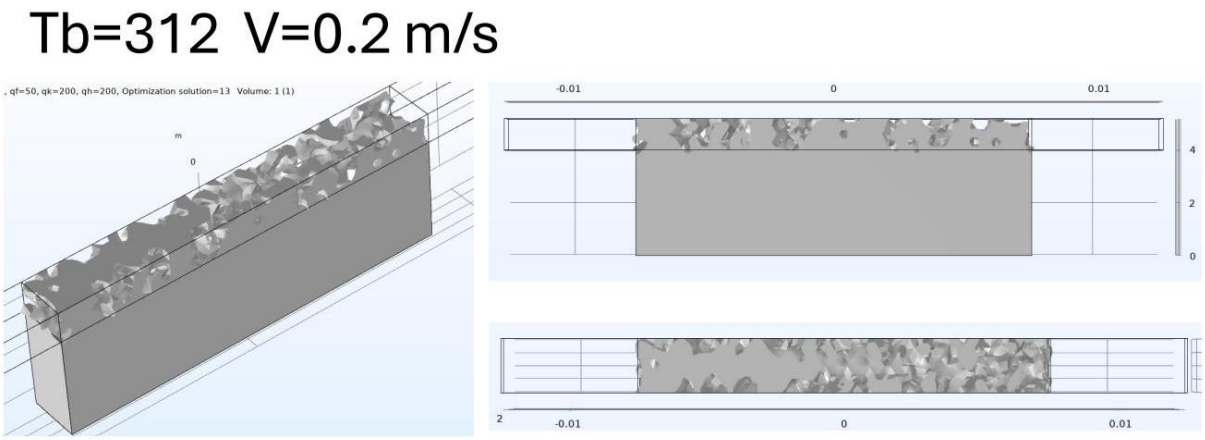


***Figure 12**. Optimized topology and results for Tb = 312 K, v_in = 0.2 m/s (tightest temperature constraint, highest flow rate).*

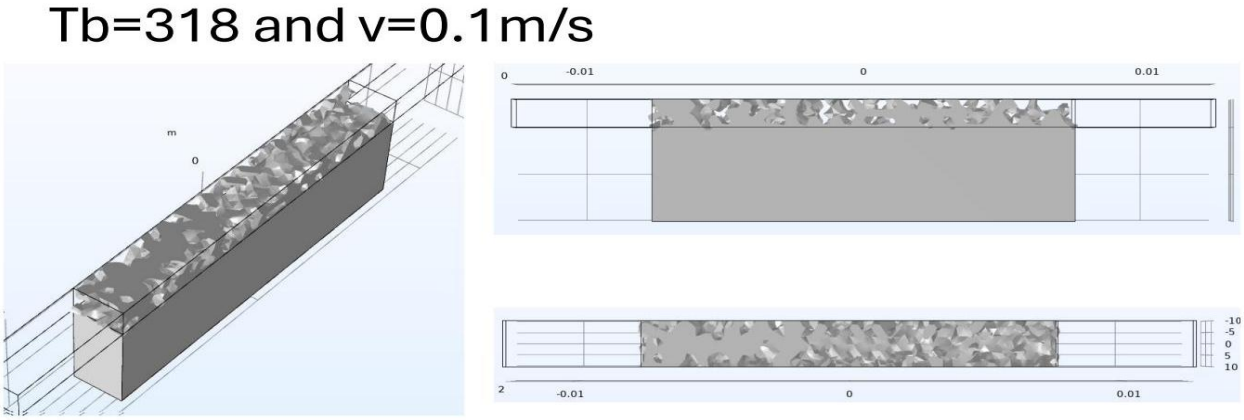


***Figure 13**. Optimized topology and results for Tb = 318 K, v_in = 0.1 m/s (baseline / reference case).*

Since the allowable base temperature is increased from 312 K to 330 K (with the corresponding reduction of the inlet velocity from 0.2 m/s to 0.05 m/s), the result is the output of progressively less densely connected, branched topologies with lower solid volume fractions due to the reduced need for cooling at higher allowable temperatures. This is consistent with physical reality since a higher allowable base temperature means that a smaller heat transfer surface area is needed (and hence pumping power too) to meet the constraint, just as seen by Marshall et al. [2] in their nine topologies.

5.6 Cross-Comparison Between the 2D and 3D Studies

Both validation studies, despite differing in dimensionality, objective sense (maximize heat transfer vs. minimize pressure drop) and material interpolation (SIMP vs. RAMP), produced optimized topologies with a common qualitative signature: material is redistributed away from a uniform fill and concentrated

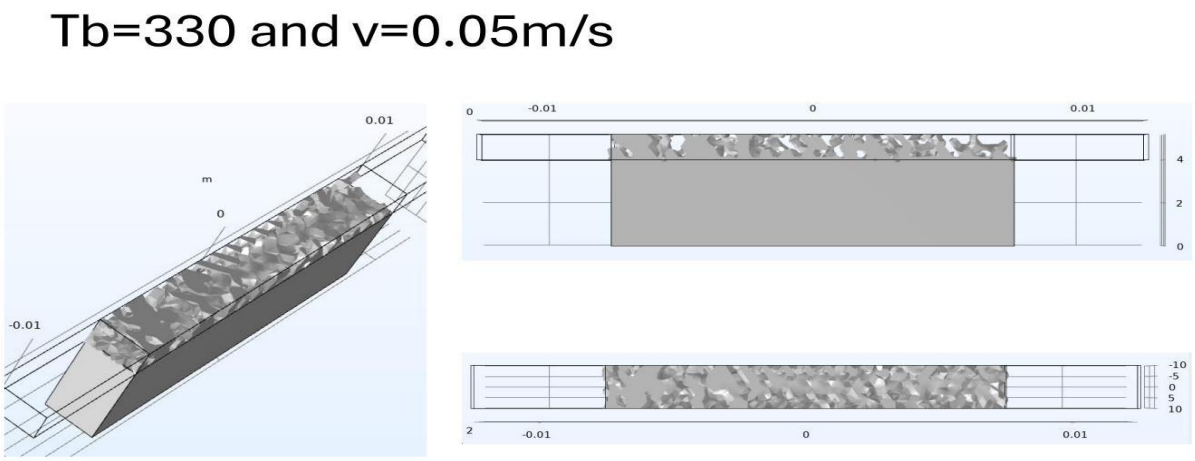


***Figure 14**. Optimized topology and results for Tb = 340 K, v_in = 0.05 m/s (baseline / reference case).*

into thin, branching, tree-like structures that balance heat-transfer area against flow resistance. This is a strong internal-consistency check — it is the expected universal signature of density-based conjugate heat transfer TO reported throughout the literature reviewed in Section 3, independent of the specific interpolation scheme or solver used. Combined with the close quantitative agreement with both benchmark studies (Section 5.1–5.5), this provides strong confidence that the governing equations, boundary conditions, material interpolation, and optimizer configuration have all been implemented correctly.

6. Conclusion

A density-based topology optimization framework for conjugate fluid–thermal systems was implemented in COMSOL Multiphysics and validated in both two and three dimensions against independent, published benchmark studies. In 2D, the framework was validated against the fin-and-tube heat exchanger optimization of Liu et al. [1] using SIMP/RAMP interpolation, reproducing both the expected convergence behavior (monotonically increasing, plateauing objective) and the characteristic branching-fin topology across a range of Reynolds numbers, heat-generation coefficients, fluids and inlet conditions. In 3D, the framework was validated against the liquid-cooled microchannel heat sink optimization of Marshall et al. [2] using a triple RAMP interpolation (permeability, conductivity, heat-transfer coefficient) and a temperature-constrained pressure-drop-minimization objective, reproducing both the qualitative dendritic channel topology and the expected pressure-drop / base-temperature trade-off across three constraint levels. Together, these results validate the correctness of the governing-equation implementation, material interpolation schemes, and optimizer configuration, and establish a verified computational pipeline ready to be extended to novel, unpublished energy-hardware design problems.

7. Future Work

Building on this validated 2D and 3D framework, the following extensions are planned for the next phase of the GRISHMA project:

- Turbulence modeling: incorporate a Brinkman-penalized turbulence closure (following Dilgen et al. [4] to extend the validated laminar framework to higher-Re industrial heat exchanger conditions.
- Multi-objective optimization: formulate a true Pareto-front optimization balancing heat-transfer performance against pumping power (following Ghasemi et al. [6]), rather than a single objective with a hard constraint.
- Manufacturing-aware constraints: introduce minimum feature size, overhang-angle and support-structure constraints so that optimized topologies are directly printable via additive manufacturing, as demonstrated by Marshall et al.'s [2] geometry-smoothing workflow.
- Benchmarking against conventional designs: quantify the efficiency gains of TO-generated geometries over conventional pin-fin arrays (Tullius et al. [7]) on a common performance metric (e.g., thermal resistance per unit pumping power).
- Application to new energy hardware: apply the validated framework to the original GRISHMA target applications — battery cooling channels, turbine-blade internal cooling passages, and offshore wind-foundation lattices — building on the passive PCM-based battery thermal management study already completed in parallel (see companion report).

8. References

1. Liu, A., Wang, G., Wang, D., Peng, X., & Yuan, H. (2021). Study on the thermal and hydraulic performance of fin-and-tube heat exchanger based on topology optimization. Applied Thermal Engineering, 197, 117380.
2. Marshall, S. D., & Lee, P. S. 3D topology optimisation of liquid-cooled microchannel heat sinks. Department of Mechanical Engineering, National University of Singapore (preprint / working paper).
3. Borrvall, T., & Petersson, J. (2003). Topology optimization of fluids in Stokes flow. International Journal for Numerical Methods in Fluids, 41(1), 77–107.
4. Dilgen, S. B., Dilgen, C. B., Fuhrman, D. R., Sigmund, O., & Lazarov, B. S. (2018). Density based topology optimization of turbulent flow heat transfer systems. Structural and Multidisciplinary Optimization, 57(5), 1905–1918.
5. Marck, G., Nemer, M., Harion, J.-L., Russeil, S., & Bougeard, D. (2012). Topology optimization using the SIMP method for multiobjective conductive problems. Numerical Heat Transfer, Part B: Fundamentals, 61(6), 439–470.
6. Ghasemi, A., & Elham, A. (2021). Multi-objective topology optimization of pin-fin heat exchangers using spectral and finite-element methods. Structural and Multidisciplinary Optimization.
7. Tullius, J. F., Tullius, T. K., & Bayazitoglu, Y. (2012). Optimization of short micro pin fins in minichannels. International Journal of Heat and Mass Transfer.
8. COMSOL, Inc. (2023–2026). COMSOL Multiphysics 6.3 Reference Manual — CFD Module, Heat Transfer Module, Optimization Module. Stockholm, Sweden.